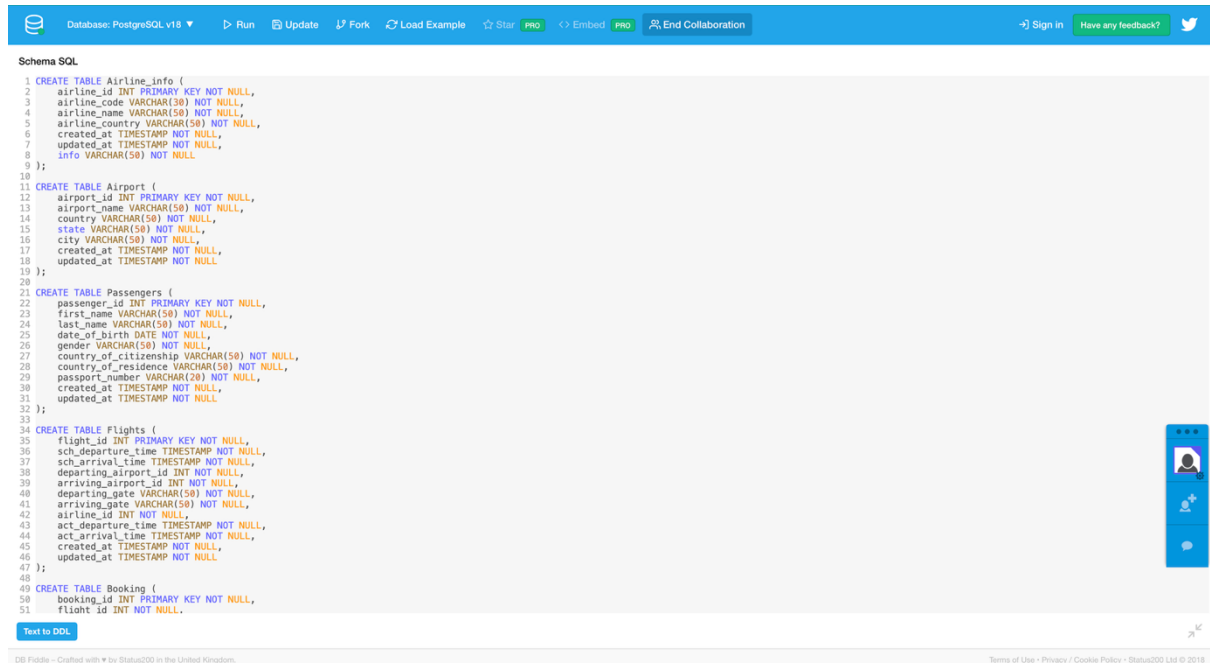


Laboratory work #2

Bakytzhan Bekassyl

Creating tables:



The screenshot shows a PostgreSQL SQL editor interface. The top bar includes the PostgreSQL logo, the database name 'Database: PostgreSQL v19', and various action buttons like 'Run', 'Update', 'Fork', 'Load Example', 'Star', 'Embed', and 'End Collaboration'. There are also links for 'Sign in' and 'Have any feedback?'. The main area is titled 'Schema SQL' and contains the following SQL code:

```
1 CREATE TABLE Airline_info (  
2     airline_id INT PRIMARY KEY NOT NULL,  
3     airline_code VARCHAR(30) NOT NULL,  
4     airline_name VARCHAR(50) NOT NULL,  
5     airline_country VARCHAR(50) NOT NULL,  
6     created_at TIMESTAMP NOT NULL,  
7     updated_at TIMESTAMP NOT NULL,  
8     info VARCHAR(50) NOT NULL  
9 );  
10  
11 CREATE TABLE Airport (  
12     airport_id INT PRIMARY KEY NOT NULL,  
13     airport_name VARCHAR(50) NOT NULL,  
14     country VARCHAR(50) NOT NULL,  
15     state VARCHAR(50) NOT NULL,  
16     city VARCHAR(50) NOT NULL,  
17     created_at TIMESTAMP NOT NULL,  
18     updated_at TIMESTAMP NOT NULL  
19 );  
20  
21 CREATE TABLE Passengers (  
22     passenger_id INT PRIMARY KEY NOT NULL,  
23     first_name VARCHAR(50) NOT NULL,  
24     last_name VARCHAR(50) NOT NULL,  
25     date_of_birth DATE NOT NULL,  
26     gender VARCHAR(50) NOT NULL,  
27     country_of_citizenship VARCHAR(50) NOT NULL,  
28     country_of_residence VARCHAR(50) NOT NULL,  
29     passport_number VARCHAR(20) NOT NULL,  
30     created_at TIMESTAMP NOT NULL,  
31     updated_at TIMESTAMP NOT NULL  
32 );  
33  
34 CREATE TABLE Flights (  
35     flight_id INT PRIMARY KEY NOT NULL,  
36     sch_departure_time TIMESTAMP NOT NULL,  
37     sch_arrival_time TIMESTAMP NOT NULL,  
38     departing_airport_id INT NOT NULL,  
39     arriving_airport_id INT NOT NULL,  
40     departing_gate VARCHAR(50) NOT NULL,  
41     arriving_gate VARCHAR(50) NOT NULL,  
42     airline_id INT NOT NULL,  
43     act_departure_time TIMESTAMP NOT NULL,  
44     act_arrival_time TIMESTAMP NOT NULL,  
45     created_at TIMESTAMP NOT NULL,  
46     updated_at TIMESTAMP NOT NULL  
47 );  
48  
49 CREATE TABLE Booking (  
50     booking_id INT PRIMARY KEY NOT NULL,  
51     flight_id INT NOT NULL,
```

At the bottom of the editor, there is a 'Text to DDL' button and a small icon of a person. The footer of the page contains the text 'DB Fiddle - Crafted with ♥ by Status200 in the United Kingdom.' and 'Terms of Use · Privacy · Cookie Policy · Status200 Ltd © 2018'.

Changes:

```
ALTER TABLE Airline_info RENAME TO Airline;  
ALTER TABLE Booking RENAME COLUMN price TO ticket_price;  
ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;  
ALTER TABLE Airline DROP COLUMN info;
```

Foreign keys:

```
104 ALTER TABLE Airline_info RENAME TO Airline;
105 ALTER TABLE Booking RENAME COLUMN price TO ticket_price;
106 ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;
107 ALTER TABLE Airline DROP COLUMN info;
108
109
110 ALTER TABLE Security_check ADD CONSTRAINT fk_seccheck_passenger
111     FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
112
113 ALTER TABLE Booking ADD CONSTRAINT fk_booking_passenger
114     FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
115
116 ALTER TABLE Baggage_check ADD CONSTRAINT fk_baggagecheck_passenger
117     FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
118
119 ALTER TABLE Baggage_check ADD CONSTRAINT fk_baggagecheck_booking
120     FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
121
122 ALTER TABLE Baggage ADD CONSTRAINT fk_baggage_booking
123     FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
124
125 ALTER TABLE Boarding_pass ADD CONSTRAINT fk_boardingpass_booking
126     FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
127
128 ALTER TABLE Booking_flight ADD CONSTRAINT fk_bookingflight_booking
129     FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
130
131 ALTER TABLE Booking_flight ADD CONSTRAINT fk_bookingflight_flight
132     FOREIGN KEY (flight_id) REFERENCES Flights(flight_id);
133
134 ALTER TABLE Flights ADD CONSTRAINT fk_flights_departing_airport
135     FOREIGN KEY (departing_airport_id) REFERENCES Airport(airport_id);
136
137 ALTER TABLE Flights ADD CONSTRAINT fk_flights_arriving_airport
138     FOREIGN KEY (arriving_airport_id) REFERENCES Airport(airport_id);
139
140 ALTER TABLE Flights ADD CONSTRAINT fk_flights_airline
141     FOREIGN KEY (airline_id) REFERENCES Airline(airline_id);
```

Text to DDL

Tables:

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Results

Query #1 Execution time: 0.24ms

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1	KC	Air Astana	Kazakhstan	2026-09-21 17:48:20.206618	2026-09-21 17:48:20.206618
2	DV	SCAT Airlines	Kazakhstan	2026-09-21 17:48:20.206618	2026-09-21 17:48:20.206618
3	LH	Lufthansa	Germany	2026-09-21 17:48:20.206618	2026-09-21 17:48:20.206618
4	TK	Turkish Airlines	Turkey	2026-09-21 17:48:20.206618	2026-09-21 17:48:20.206618
5	EK	Emirates	UAE	2026-09-21 17:48:20.206618	2026-09-21 17:48:20.206618

Query #2 Execution time: 0.19ms

airport_id	airport_name	country	state	city	created_at	updated_at
1	Almaty International Airport	Kazakhstan	Almaty	Almaty	2026-09-21 17:48:20.207136	2026-09-21 17:48:20.207136
2	Astana International Airport	Kazakhstan	Astana	Astana	2026-09-21 17:48:20.207136	2026-09-21 17:48:20.207136
3	Frankfurt Airport	Germany	Hesse	Frankfurt	2026-09-21 17:48:20.207136	2026-09-21 17:48:20.207136
4	Istanbul Airport	Turkey	Istanbul	Istanbul	2026-09-21 17:48:20.207136	2026-09-21 17:48:20.207136
5	Dubai International Airport	UAE	Dubai	Dubai	2026-09-21 17:48:20.207136	2026-09-21 17:48:20.207136

Query #3 Execution time: 0.24ms

passenger_id	first_name	last_name	date_of_birth	gender	country_of_citizenship	country_of_residence	passport_number	created_at	updated_at
1	Ali	Saparov	2000-01-15	Male	Kazakhstan	Kazakhstan	KZ123456	2026-09-21 17:48:20.207613	2026-09-21 17:48:20.207613
2	John	Smith	1999-05-20	Male	USA	USA	US123456	2026-09-21 17:48:20.207613	2026-09-21 17:48:20.207613

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Results

Query #3 Execution time: 0.24ms

passenger_id	first_name	last_name	date_of_birth	gender	country_of_citizenship	country_of_residence	passport_number	created_at	updated_at
1	Ali	Saparov	2000-01-15	Male	Kazakhstan	Kazakhstan	KZ123456	2026-09-21 17:48:20.207613	2026-09-21 17:48:20.207613
2	John	Smith	1999-05-20	Male	USA	USA	US123456	2026-09-21 17:48:20.207613	2026-09-21 17:48:20.207613
3	Anna	Brown	2001-03-10	Female	Germany	Germany	DE123456	2026-09-21 17:48:20.207613	2026-09-21 17:48:20.207613
4	Algerim	Nurgali	2002-07-12	Female	Kazakhstan	Kazakhstan	KZ654321	2026-09-21 17:48:20.207613	2026-09-21 17:48:20.207613
5	Mehmet	Kaya	1998-11-25	Male	Turkey	Turkey	TR123456	2026-09-21 17:48:20.207613	2026-09-21 17:48:20.207613

Query #4 Execution time: 0.2ms

flight_id	sch_departure_time	sch_arrival_time	departing_airport_id	arriving_airport_id	departing_gate	arriving_gate	airline_id	act_departure_time	act_arrival_time	created_at	updated_at
1	2026-09-22 08:00:00	2026-09-22 10:00:00	1	2	A1	B1	1	2026-09-22 08:10:00	2026-09-22 10:05:00	2026-09-21 17:48:20.20815	2026-09-21 17:48:20.20815
2	2026-09-22 12:00:00	2026-09-22 16:00:00	2	3	A2	B2	3	2026-09-22 12:05:00	2026-09-22 16:10:00	2026-09-21 17:48:20.20815	2026-09-21 17:48:20.20815
3	2026-09-23 09:00:00	2026-09-23 13:00:00	1	4	A3	B3	4	2026-09-23 09:00:00	2026-09-23 13:05:00	2026-09-21 17:48:20.20815	2026-09-21 17:48:20.20815
4	2026-09-24 14:00:00	2026-09-24 18:00:00	4	5	A4	B4	5	2026-09-24 14:10:00	2026-09-24 18:20:00	2026-09-21 17:48:20.20815	2026-09-21 17:48:20.20815
5	2026-09-25 16:00:00	2026-09-25 20:00:00	5	1	A5	B5	2	2026-09-25 16:05:00	2026-09-25 20:10:00	2026-09-21 17:48:20.20815	2026-09-21 17:48:20.20815

Query #5 Execution time: 0.19ms

booking_id	flight_id	passenger_id	booking_platform	created_at	updated_at	status	ticket_price
1	1	1	Website	2026-09-21 17:48:20.209185	2026-09-21 17:48:20.209185	Confirmed	120.00

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Query #5Execution time: 0.19ms

booking_id	flight_id	passenger_id	booking_platform	created_at	updated_at	status	ticket_price
1	1	1	Website	2026-09-21 17:48:20.209185	2026-09-21 17:48:20.209185	Confirmed	120.00
2	2	2	Mobile App	2026-09-21 17:48:20.209185	2026-09-21 17:48:20.209185	Confirmed	350.00
3	3	3	Website	2026-09-21 17:48:20.209185	2026-09-21 17:48:20.209185	Pending	280.00
4	4	4	Agent	2026-09-21 17:48:20.209185	2026-09-21 17:48:20.209185	Confirmed	450.00
5	5	5	Website	2026-09-21 17:48:20.209185	2026-09-21 17:48:20.209185	Confirmed	500.00

Query #6Execution time: 0.19ms

baggage_id	weight_in_kg	created_at	updated_at	booking_id
1	20.50	2026-09-21 17:48:20.209843	2026-09-21 17:48:20.209843	1
2	18.00	2026-09-21 17:48:20.209843	2026-09-21 17:48:20.209843	2
3	22.30	2026-09-21 17:48:20.209843	2026-09-21 17:48:20.209843	3
4	15.50	2026-09-21 17:48:20.209843	2026-09-21 17:48:20.209843	4
5	25.00	2026-09-21 17:48:20.209843	2026-09-21 17:48:20.209843	5

Query #7Execution time: 0.16ms

boarding_pass_id	booking_id	seat	boarding_time	created_at	updated_at
1	1	12A	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187

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Query #7Execution time: 0.16ms

boarding_pass_id	booking_id	seat	boarding_time	created_at	updated_at
1	1	12A	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187
2	2	14B	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187
3	3	15C	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187
4	4	16D	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187
5	5	17E	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187	2026-09-21 17:48:20.220187

Query #8Execution time: 0.16ms

booking_flight_id	booking_id	flight_id	created_at	updated_at
1	1	1	2026-09-21 17:48:20.222455	2026-09-21 17:48:20.222455
2	2	2	2026-09-21 17:48:20.222455	2026-09-21 17:48:20.222455
3	3	3	2026-09-21 17:48:20.222455	2026-09-21 17:48:20.222455
4	4	4	2026-09-21 17:48:20.222455	2026-09-21 17:48:20.222455
5	5	5	2026-09-21 17:48:20.222455	2026-09-21 17:48:20.222455

Query #9Execution time: 0.16ms

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
------------------	--------------	------------	------------	------------	--------------

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3	3	3	2026-09-21 17:48:20.222455	2026-09-21 17:48:20.222455		
4	4	4	2026-09-21 17:48:20.222455	2026-09-21 17:48:20.222455		
5	5	5	2026-09-21 17:48:20.222455	2026-09-21 17:48:20.222455		

Query #9 Execution time: 0.16ms

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
1	Passed	2026-09-21 17:48:20.2216	2026-09-21 17:48:20.2216	1	1
2	Passed	2026-09-21 17:48:20.2216	2026-09-21 17:48:20.2216	2	2
3	Passed	2026-09-21 17:48:20.2216	2026-09-21 17:48:20.2216	3	3
4	Passed	2026-09-21 17:48:20.2216	2026-09-21 17:48:20.2216	4	4
5	Passed	2026-09-21 17:48:20.2216	2026-09-21 17:48:20.2216	5	5

Query #10 Execution time: 0.16ms

security_check_id	check_result	created_at	updated_at	passenger_id
1	Passed	2026-09-21 17:48:20.220977	2026-09-21 17:48:20.220977	1
2	Passed	2026-09-21 17:48:20.220977	2026-09-21 17:48:20.220977	2
3	Passed	2026-09-21 17:48:20.220977	2026-09-21 17:48:20.220977	3
4	Passed	2026-09-21 17:48:20.220977	2026-09-21 17:48:20.220977	4
5	Passed	2026-09-21 17:48:20.220977	2026-09-21 17:48:20.220977	5

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1NF: The table violates 1NF because the seat_numbers column contains multiple values in one cell. Each field should contain only one atomic value.

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Fiddle Description

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Schema SQL

```
1 CREATE TABLE booking_receipt (  
2   booking_id INT,  
3   passenger_full_name VARCHAR(100),  
4   passenger_passport_number VARCHAR(20),  
5   flight_number VARCHAR(20),  
6   departure_airport_name VARCHAR(100),  
7   departure_city VARCHAR(50),  
8   arrival_airport_name VARCHAR(100),  
9   arrival_city VARCHAR(50),  
10  airline_name VARCHAR(100),  
11  seat_numbers VARCHAR(100),  
12  ticket_price DECIMAL(10,2)  
13 );  
14  
15 INSERT INTO booking_receipt VALUES  
16 (1, 'Ali Saparov', 'KZ123456', 'KC101',  
17 'Almaty International Airport', 'Almaty',  
18 'Astana International Airport', 'Astana',  
19 'Air Astana', '12A, 12B', 120.00),  
20  
21 (2, 'John Smith', 'US123456', 'LH202',  
22 'Astana International Airport', 'Astana',
```

Text to DDL

Query SQL

```
1 SELECT * FROM booking_receipt;
```

Results

Copy as Markdown

booking_id	passenger_full_name	passenger_passport_number	flight_number	departure_airport_name	departure_city	arrival_airport_name	arrival_city	airline_name	seat_numbers	ticket_price
1	Ali Saparov	KZ123456	KC101	Almaty International Airport	Almaty	Astana International Airport	Astana	Air Astana	12A, 12B	120.00
2	John Smith	US123456	LH202	Astana International Airport	Astana	Frankfurt Airport	Frankfurt	Lufthansa	14A	350.00

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INF: The table is now in 1NF because each field contains only one atomic value. The seat_numbers column was changed to seat_number, and each seat is stored in a separate row. The primary key is (booking_id, seat_number).

Database: PostgreSQL v13

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Schema SQL

```
9 arrival_city VARCHAR(50),
10 airline_name VARCHAR(100),
11 seat_numbers VARCHAR(100),
12 ticket_price DECIMAL(10,2)
13 ;
14
15 INSERT INTO booking_receipt VALUES
16 (1, 'Ali Saparov', 'KZ123456', 'KC101',
17 'Almaty International Airport', 'Almaty',
18 'Astana International Airport', 'Astana',
19 'Air Astana', '12A, 12B', '120.00'),
20
21 (2, 'John Smith', 'US123456', 'LH202',
22 'Astana International Airport', 'Astana',
23 'Frankfurt Airport', 'Frankfurt',
24 'Lufthansa', '14A', '350.00');
25
26 DROP TABLE IF EXISTS booking_receipt_Inf;
27
28 CREATE TABLE booking_receipt_Inf (
29 booking_id INT,
30 passenger_full_name VARCHAR(100),
31 passenger_passport_number VARCHAR(70))
```

Text to DDL

Query SQL

```
1
2 SELECT * FROM booking_receipt_Inf;
```

Results

Copy as Markdown

Query #1 Execution time: 0.19ms

booking_id	passenger_full_name	passenger_passport_number	flight_number	departure_airport_name	departure_city	arrival_airport_name	arrival_city	airline_name	seat_number	ticket_price
1	Ali Saparov	KZ123456	KC101	Almaty International Airport	Almaty	Astana International Airport	Astana	Air Astana	12A	120.00
1	Ali Saparov	KZ123456	KC101	Almaty International Airport	Almaty	Astana International Airport	Astana	Air Astana	12B	120.00
2	John Smith	US123456	LH202	Astana International Airport	Astana	Frankfurt Airport	Frankfurt	Lufthansa	14A	350.00

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2NF: The tables are now in 2NF. The attributes that depend only on booking_id were separated from the seat information. The booking_seat_2nf table uses (booking_id, seat_number) as a composite primary key.

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Schema SQL

```
49 (1, 'Ali Saparov', 'KZ123456', 'KC101',
50 'Almaty International Airport', 'Almaty',
51 'Astana International Airport', 'Astana',
52 'Air Astana', '12B', 120.00),
53
54 (2, 'John Smith', 'US123456', 'LH202',
55 'Astana International Airport', 'Astana',
56 'Frankfurt Airport', 'Frankfurt',
57 'Lufthansa', '14A', 350.00);
58
59 DROP TABLE IF EXISTS booking_2nf;
60 DROP TABLE IF EXISTS booking_seat_2nf;
61
62 CREATE TABLE booking_2nf (
63   booking_id INT PRIMARY KEY,
64   passenger_full_name VARCHAR(100),
65   passenger_passport_number VARCHAR(20),
66   flight_number VARCHAR(20),
67   departure_airport_name VARCHAR(100),
68   departure_city VARCHAR(50),
69   arrival_airport_name VARCHAR(100),
70   arrival_city VARCHAR(50),
71   airline_name VARCHAR(100),
72   ticket_price DECIMAL(10,2);
73 );
```

Query SQL

```
1 SELECT * FROM booking_2nf;
2 SELECT * FROM booking_seat_2nf;
```

Results

booking_id	passenger_full_name	passenger_passport_number	flight_number	departure_airport_name	departure_city	arrival_airport_name	arrival_city	airline_name	ticket_price
1	Ali Saparov	KZ123456	KC101	Almaty International Airport	Almaty	Astana International Airport	Astana	Air Astana	120.00
2	John Smith	US123456	LH202	Astana International Airport	Astana	Frankfurt Airport	Frankfurt	Lufthansa	350.00

Query #2

Execution time: 0.16ms

booking_id	seat_number
1	12A
1	12B
2	14A

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